

Dr. Kornpob Bhirombhakdi

Interdisciplinary Researcher & Educator

- Dual Ph.D.s (Physics + Economics)
 - Postdoctoral Astrophysicist at STScI (AURA/NASA)
 - Independent AI jailbreak and vulnerability researcher
 - Speaker at *2600TH Red x Blue Pill 2026*
 - Winner of *The Judgement Day 2026 Phase-1 Scenario Submission* Scenario: Counter-UAS System — Friendly Fire Decision
 - Research interests: human-AI collaboration, art-science integration, AI literacy, AI ethics, AI governance, AI safety, quantum computing
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Links

- bkornpob@gmail.com
 - kbhirombhakdi.weebly.com
 - bkornpob.github.io
 - ORCID: 0000-0003-0136-1281
 - linkedin.com/in/bkornpob
 - IMDb
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Education

- **Postdoc Astrophysics** | Space Telescope Science Institute (AURA/NASA), Baltimore, MD USA | 2019-2024
 - Spectroscopic studies of cosmic explosions
 - Multimessenger studies of gravitational-wave sources
 - **Ph.D. Physics** | Ohio University, Athens, OH USA | 2014–2019
 - Light Curve Powering Mechanism of Superluminous Supernovae
 - **M.S. Physics** | Ohio University, Athens, OH USA | 2014–2015
 - Project: Theoretical Studies on Nanophysics Material Properties within Complex Dielectric Environments and Multiphase Configurations
 - Recipient of Graduate Awards
 - **Physics Pre-Degree** | Ramkhamhaeng University, Bangkok TH | 2013–2014
 - Project: Independent Study on General Relativity Theory
 - **Ph.D. Economics** | Chulalongkorn University, Bangkok TH | 2009–2013
 - Mechanism Design and Game Theory + Experimental Economics
 - Studies on reciprocity and cooperative-decision drives
 - Recipient of The 90th Anniversary of Chulalongkorn University Fund
 - **M.S. Health Economics** | Chulalongkorn University, Bangkok TH | 2007–2009
 - Technical Efficiency of University Hospitals
 - **B.S. Medical Science** | Siriraj Hospital, Mahidol University, Bangkok TH | 2003–2007
 - Second Class Honours
 - Recipient of Medical Scholarship Program
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Certifications and Training

- **Quantum Computing**, IBM Quantum Platform, 2026
 - **Global MOOC on the Ethics of AI**, unesco, 2026
 - **Certified in Cybersecurity**, ISC2, 2026 – 2028
 - **Cybersecurity Trainings**, THNCA+NCSA Thailand, 2025
 - **Cybersecurity Professional Course**, NCSA Thailand, 2025
 - **Basic Cybersecurity**, NCSA Thailand, 2025
 - **Deep Learning Specialization**, DeepLearning.AI (Coursera), 2025
 - **Data Science and Machine Learning Specialization**, Johns Hopkins University (Coursera), 2015 – 2017
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Employment

- **AI Systems Architect & Technical Advisor** | DR.KB (Self-Employed), Bangkok, Thailand | Sep 2025 – Present Offer end-to-end execution across AI problems, solutions, and integrations. Consult C-suite leaders and product teams on enterprise strategy, risk mitigation, and human-AI workflows, while collaborating closely with grassroots user communities, developers, and cross-functional learners across all career stages. Train teams on secure AI agent architectures and AI literacy, design custom machine learning and LLM pipelines, and rigorously evaluate system safety, vulnerability, and robustness for diverse audiences.
- **Lead Prompt Security Researcher** | InjectPrompt.com (Contract), Remote | Jan 2026 – May 2026 Led red-team research on AI agent safety and jailbreak tactics. Designed custom RAG knowledge bases, skills, and orchestration workflows for sandboxed jailbreak testing. Authored and released *InjectPrompt Companion 3.0* and the *Jailbreak Perturbation Framework (JPF)*.
- **Data Scientist** | Sertis Co., Ltd., Bangkok, Thailand | Aug 2024 – Aug 2025
 - Developed deep learning architectures for computer vision and multi-object tracking.
 - Built scalable multi-branch forecasting models for retail and wholesale supply chains.
 - Maintained and optimized enterprise-grade ML assets.
 - Delivered expert consulting and pitched high-value extension solutions.
- **Postdoctoral Researcher** | Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA | Aug 2019 – Feb 2024
 - Investigated the physics and systemic impacts of cosmic explosions, including supernovae and gamma-ray bursts (GRBs).
 - Spearheaded successful observational proposals utilizing both the Hubble Space Telescope (HST) and James Webb Space Telescope (JWST).
 - Collaborated globally as an active member of the ENGRAVE collaboration tracking gravitational-wave and cosmic transient counterparts.
 - Served in multiple institutional and community roles supporting the STScI research ecosystem and broader astronomy community.
- **Research/Teaching Assistant** | Physics and Astronomy Department, Ohio University, Athens, OH, USA | Aug 2014 – May 2019
 - Spearheaded observational astrophysics research on superluminous supernovae, light curve powering mechanisms, and transient data analysis.
 - Instructed, and evaluated undergraduate physics laboratory coursework, fostering rigorous experimental methodology and data rigor.

- Directed physics homework help-room sessions, mentoring students through complex mechanics, electromagnetism, and calculus-based problem solving.
 - Coordinated grading, evaluation, and proctoring frameworks for large-enrollment undergraduate physics courses.
- **Senior Consultant** | LEARN Corporation Co., Ltd., Bangkok, Thailand | Jan 2012 – Jan 2014
 - Pioneered early data transformation initiatives at the onset of the big-data analytics era, preceding the modern machine learning boom.
 - Spearheaded data-driven market research and product improvement strategies, translating complex user metrics into actionable business insights.
 - Designed analytical frameworks to transition legacy educational and business products into modern data-centric operations.
- **Market Research Consultant** | Freelance, Bangkok, Thailand | Jan 2010 – Jan 2012
 - Conducted independent quantitative and qualitative market research studies, analyzing consumer trends and competitive landscapes for diverse commercial clients.
 - Synthesized strategic research data into actionable market entry and product positioning reports.
- **Coordinator Assistant** | Economic Impacts of Demographic Change Project, Thailand Research Fund, Bangkok, Thailand | Jan 2011 – Jan 2013
 - Partnered directly with the project coordinator to manage large-scale administrative operations, cross-institutional communications, and researcher networking.
 - Designed meeting agendas, coordinated multi-stakeholder workshops, and streamlined operational workflows across research groups.
 - Managed the editorial production and publishing pipelines for annual newsletters and comprehensive project reports.
- **Lecturer** | 2010, 2012 | Faculty of Hospitality and Tourism, Prince of Songkla University, Phuket, Thailand
 - Principles of Economics (Undergraduate Level; International Program)
- **Lecturer** | 2009 | Faculty of Economics, Khonkaen University, Nongkhai, Thailand
 - Health Economics (Undergraduate Level)
- **Mathematician** | Institute of Academic Development Inc., Bangkok, Thailand | 2009
 - Executed rigorous technical proofreading and verification of advanced mathematical curricula and exercise sets.
 - Audited problem solutions to ensure absolute logical, numerical, and methodological correctness before publication.
- **Teaching Assistant** | Faculty of Economics, Chulalongkorn University, Bangkok, Thailand
 - Advanced Microeconomic Theory II (Ph.D. Level; International Program) — 2010, 2011
 - Statistics for Health Economics (Graduate Level; International Program) — 2011
 - Introduction to Econometrics (Undergraduate Level) — 2010 – 2012
 - Introduction to Economics (Undergraduate Level) — 2010
 - Math for Economist I (Undergraduate Level) — 2009
 - Math for Economist II (Undergraduate Level) — 2012
- **Consultant for Business Research and Development** | Freelance, Bangkok, Thailand | 2007 – 2014
 - Partnered with diverse clients to deliver comprehensive market research, data-driven consumer insights, and competitive landscape evaluations.
 - Designed quantitative and qualitative methodologies to optimize product strategies, validate business models, and shape early analytics frameworks.

- **Tutor** | Freelance, Bangkok, Thailand | 2007 – 2014
 - Delivered private instruction and large-scale national exam preparation classes sponsored by corporate and institutional partners.
 - Taught advanced multidisciplinary curricula spanning mathematics, physics, economics, statistics, and calculus across high school, undergraduate, and graduate levels.
 - **Actor/Model/DJ/Brand Ambassador** | Freelance, Bangkok, Thailand | 2005 – 2010
 - IMDb
 - Starred and performed across commercial media, television, and modeling campaigns, including credited appearances in *Club M2* and *Love Audition*.
 - Served as a professional DJ and brand ambassador for high-profile lifestyle and entertainment events across Bangkok.
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Software and Applications

- **InjectPrompt Companion 3.0** | companion.injectprompt.com | Structured prompt engineering and systematic extraction methodologies for safety testing. Multi-tactic jailbreak templates following the Jailbreak Perturbation Framework (JPF). A progressive learning curriculum powering research from beginner through advanced levels.
 - **hstgrism** | pypi.org/project/hstgrism/ | Python 3 package for reducing HST grism observations, providing a modular, aXe-compatible workflow with simpler installation, diagnostics, and flexible point-source extraction. Optimized for flexible point-source extraction and 2D polynomial background modeling. Powering data reduction pipelines in peer-reviewed observational studies, including Bhirombhakdi et al. (2024).
 - **EchoEcho** | github.com/OhioUniversityGameDevelopersAssociation/EchoEcho | Help Fernando the Bat echolocate the way out and survive attacks from mad stone gargoyles. A prototype from Global Game Jam 2017, together with the Ohio University Game Developers Association (OUGDA), exploring wave and darkness.
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Achievements

- **Winner, The Judgement Day — Call for Scenario Challenge (Track 1)** | 2026
Scenario: Counter-UAS System — Friendly Fire Decision (IFF Corruption Attack). [Link](#)
- **Space Telescope Proposals** | 2024
 - with the *Hubble Space Telescope* 14 proposals
 - with the *James Webb Space Telescope* 6 proposals
- **Bravo! Awards** | Space Telescope Science Institute (AURA/NASA)
 - As STScI scientist, hosting the Loyola University Physics Club on February 20, 2020
 - As subject-matter expert, contributing in Astrogrism Sprint during May 12-21, 2020
- **Graduate Awards** | 2014 – 2019 Ohio University, Athens, OH, USA
- **Global Game Jam 2017** | 2017 Lead Programmer and Project Coordinator in the project “EchoEcho” with a team at GRID Lab, Ohio University, Athens, OH, USA. [Link](#)
- **Kendo 2-Dan** | 2017 All United States Kendo Federation, USA (AUSKF ID: 017666)
- **Executive Course on Social Security** | 2012 As subject-matter expert, representative from the Faculty of Economics, Chulalongkorn University, organized by the International Labour Organization (UN), in collaboration with Chulalongkorn University, Bangkok, Thailand

- **Hands-on Training “How to Cost, Finance and Monitor Social Protection Schemes?”** | 2012 As subject-matter expert, representative from the Faculty of Economics, Chulalongkorn University, organized by the International Labour Organization (UN), in collaboration with Chulalongkorn University and Health Insurance Research Institute of Thailand, Bangkok, Thailand
 - **The Best Paper Awards** | 2012 “Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model” at The 6th Asian Business Research Conference, 8–10 April 2012, Bangkok, Thailand.
 - **Associate Fellowship** | 2012 World Academy of Social Science, Victoria, Australia
 - **Research Grants** | 2012 The 90th Anniversary of Chulalongkorn University Fund (Ratchadaphiseksomphot Endowment Fund), Chulalongkorn University, Bangkok, Thailand
 - **Brand Ambassador** | 2008 Playsmart Idol, organized by NC True Co., Ltd., Thailand. [Link](#)
 - **Medical Scholarship Program** | 2004 – 2013 Mahidol University, Bangkok, Thailand
 - **Gold Medal** | 2002 National Physics Olympiad Competition, The Promotion of Academic Olympiads, Thailand
 - **High Distinction Awards** | 2001 Australian National Chemistry Quiz, senior division, Royal Australian Chemical Institute
 - **Thailand Representative** | 2001 Arirang Youth Camp 5th, UNESCO, South Korea
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Services

- **Independent AI Security and Red-Teaming Researcher and Contributor** | 2025 – Present
 - *BASI*: Research contributor sharing advanced jailbreak payloads, red-teaming methodologies, and collaborative notes with the elite adversary community.
 - *2600TH Red x Blue Pill 2026*: Speaker
- **Member of the Electromagnetic counterparts of gravitational wave sources at the Very Large Telescope (ENGRAVE)** | 2019 - 2024 International collaboration bringing together astronomers who use the ESO facilities to research gravitational wave events. [Link](#)
- **Member of Slitless Spectroscopy Group** | 2019 - 2024 Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Member of Review Editors** | 2023 *Frontiers in Astronomy and Space Sciences*. [Link](#)
- **Representative of Postdoctoral Fellows** | 2019 - 2023 Research Computing Forum at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Scientific Collaborator** | *Hearing Stars - SN1054 (Supernova)* by Seo Yoon Soyoona Kim | STScI 2021
 - Provided astrophysical framework, historical Supernova 1054 archival data context, and scientific insights for the multimedia composition and cosmic soundscape inspired by the Crab Nebula remnant.
- **Journal Club Organizer (Remote)** | 2020 - 2021 The Compact Objects and Supernovae Journal Club at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Leveler** | 2021 Hubble Space Telescope Cycle 29’s time allocation process. [Link](#)
- **Subject-Matter Expert** | 2020 Astrogrism Kickoff sprint at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA. [Link](#)
- **Member of the Host** | 2019 Visit by the Loyola University Physics Club at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA

- **Member of the Judges** | 2019 Chambliss Astronomy Achievement Student Awards at the American Astronomical Society, AAS Meeting #233
 - **Member of the Judges** | 2018 Science Fair at the West Elementary, Athens, OH, USA
 - **Proposal Reviewer Committee** | 2018 Original Work Grant and Travel Awards, in collaboration with The Graduate College and the Graduate Student Senate at Ohio University, Athens, OH, USA
 - **Kendo Demonstration** | 2014 – 2018 Various community events, in collaboration with the Ohio University Kendo Club, Ohio University, Athens, OH, USA
 - **LGBT Affairs Commissioner** | 2017 Graduate Student Senate, Ohio University, Athens, OH, USA
 - **Science Staff** | 2015, 2017 Physics and Astronomy Open House organized by the Physics and Astronomy Department, Ohio University, Athens, OH, USA
 - **Volunteer** | 2016 Athens Beautification Day, Athens, OH, USA
 - **Science Staff** | 2014 Ohio Valley Museum of Discovery, Athens, OH, USA
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Publications and Talks

- **An Afterglow Study of the “New Year’s Burst” GRB 220101A** | with Roychowdhury et al. (In Progress) arXiv:2602.04660. [Link](#)
- **Unchained Prompts** | Bhirombhakdi, K. (In Progress) Speculative art for AI security & public literacy.
- **Judgement Day: An Anatomy of Successful Multimodal Attacks on Safety-Critical {AI} Systems** | with Choi et al. (2026) ICLR 2027 Conference Submission. [OpenReview](#)
- **Talk: jailbreaks, agent vulnerabilities, and the grey mage tower** | Bhirombhakdi, K. (2026) 2600TH Red x Blue Pill 2026, Bangkok, Thailand. [github](#)
- **Cyberpunk Redteam Fable Zines & Audios Series** | Bhirombhakdi, K. (2026) An open-access research series exploring human-AI collaboration, AI literacy, ethics, governance, safety. Demonstrate an experimental art-science synergy by integrating cyberpunk-redteam-fables narratives with technical research on AI jailbreaks, scheming, and defense to deliver accessible, high-impact cybersecurity education. Examples: i) ZADDY Cooking Club · audios — Translating complex AI jailbreaks into narrative-driven technical writeups to enhance non-technical AI literacy. ii) ZADDY Breadcrumbs 2026 Collection · audios — 20+ breadcrumb notes exploring the quickly evolving AI development and emerging threats.
- **Jailbreak Perturbation Framework: Theory and Applications** | Bhirombhakdi, K., Willis-Owen, D., and N, K. (2026) White paper exploring a unified framework for jailbreak search and mutation, and its implementation on the InjectPrompt Companion 3.0. [zenodo:20040972](#)
- **Probing the distant universe through GRB afterglow spectroscopy** | with Tanvir et al. (2025) JWST Proposal, Cycle 4, ID. #8539. [Link](#)
- **The Redshift of GRB 190829A/SN 2019oyw: A Case Study of GRB-SN Evolution** | Bhirombhakdi et al. (2024) The Astrophysical Journal, Volume 977, Issue 2, id.256, 16 pp. [Link](#)
- **A Hubble Space Telescope Search for r-Process Nucleosynthesis in Gamma-ray Burst Supernovae** | with Rastinejad et al. (2024) The Astrophysical Journal, Volume 968, Issue 1, id.14, 17 pp. [Link](#)

- **Heavy-element Production in a Compact Object Merger Observed by JWST** | with Levan et al. (2024) *Nature*, Volume 626, Issue 8000, p. 737–741. [Link](#)
- **A Long-duration Gamma-ray Burst of Dynamical Origin from the Nucleus of an Ancient Galaxy** | with Levan et al. (2023) *Nature Astronomy*, Volume 7, p. 976–985. [Link](#)
- **Panning for Gold, but Finding Helium: Discovery of the Ultra-stripped Supernova SN2019wxt from Gravitational-wave Follow-up Observations** | with EN-GRAVE Collaboration (2023) *Astronomy and Astrophysics*, Volume 675, id.A201, 34 pp. [Link](#)
- **The First JWST Spectrum of a GRB Afterglow: No Bright Supernova in Observations of the Brightest GRB of all Time, GRB 221009A** | with Levan et al. (2023) *The Astrophysical Journal Letters*, Volume 946, Issue 1, id.L28, 14 pp. [Link](#)
- **An Extremely Energetic Supernova from a Very Massive Star in a Dense Medium** | with Nicholl et al. (2020) *Nature Astronomy*, Volume 4, p. 893–899. [Link](#)
- **Observational Constraints on the Optical and Near-infrared Emission from the Neutron Star-Black Hole Binary Merger Candidate S190814bv** | with Ackley et al. (2020) *Astronomy and Astrophysics*, Volume 643, id.A113, 48 pp. [Link](#)
- **SN2008es: Strong Interacting Hydrogen-rich Superluminous Supernova Without Narrow H-alpha Features and Early Dust Formation** | Bhirombhakdi (2019) *STScI Newsletters*, Volume 36, Issue 03. [Link](#)
- **The Type II Superluminous SN2008es at Late Times: Near-infrared Excess and Circumstellar Interaction** | Bhirombhakdi et al. (2019) *Monthly Notices of the Royal Astronomical Society*, Volume 488, Issue 3, p. 3783–3793. [Link](#)
- **Light Curve Powering Mechanisms of Superluminous Supernovae** | Bhirombhakdi (2019) Dissertation, Ohio University, Athens, OH, USA. [Link](#)
- **Talk: Hunting Magnetar Central Engines in Superluminous Supernovae** | (2019) American Astronomical Society, AAS Meeting #233, id.410.04. [Link](#)
- **Where is the Engine Hiding its Missing Energy? Constraints from a Deep X-ray Non-detection of Superluminous SN2015bn** | Bhirombhakdi et al. (2018) *The Astrophysical Journal Letters*, Volume 868, Issue 2, article id. L32, 7 pp. [Link](#)
- **One Thousand Days of SN2015bn: HST Imaging Shows a Light Curve Flattening Consistent with Magnetar Predictions** | with Nicholl et al. (2018) *The Astrophysical Journal Letters*, Volume 866, Issue 2, article id. L24, 7 pp. [Link](#)
- **The Type I Superluminous Supernova PS16aqv: Lightcurve Complexity and Deep Limits on Radioactive Ejecta in a Fast Event** | with Blanchard et al. (2018) *The Astrophysical Journal*, Volume 865, Issue 1, article id. 9, 17 pp. [Link](#)
- **Spectroscopic Classification of Two Superluminous Supernovae** | with Chornock et al. (2016) *The Astronomer's Telegram* #8790. [Link](#)
- **Essays in Experimental Studies on Positive Reciprocity and Auction Design for an Object with Countervailing Positive Externalities** | Bhirombhakdi (2013) Dissertation, Chulalongkorn University, Bangkok, Thailand. [Link](#)
- **Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model** | Bhirombhakdi and Potipiti (2012) *Journal of Business and Policy Research*, Volume 7, Number 3, September 2012 Special Issue, ISSN: 1449-387X. [Link to preprint version](#)
- **Talk: Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model** | (2012) The 6th Asian Business Research Conference, Bangkok, Thailand
- **Performance of a Reciprocity Model in Predicting a Positive Reciprocity Decision** | Bhirombhakdi and Potipiti (2011) *Chulalongkorn Journal of Economics*, Vol. 23, No. 1. [Link](#)
- **Talk: Performance of a Reciprocity Model in Predicting a Positive Reciprocity Decision** | (2011) International Conference on Management, Economics and Social Sciences

(ICMESS'2011), Bangkok, Thailand

- **Talk: Neuroeconomics** | (2010, 2011) Faculty of Economics, Chulalongkorn University, Bangkok, Thailand
- **Special Lecture: Opportunity Cost and Economic Evaluation** | (2010) The College of Public Health Science, Chulalongkorn University, Bangkok, Thailand
- **Special Lecture: Efficiency Analysis** | (2010) The College of Public Health Science, Chulalongkorn University, Bangkok, Thailand
- **Special Lecture: National Examination Preparation (Physics)** | (2009) (in Thai) organized by Buriram Provincial Administration Organization, Buriram, Thailand
- **Special Lecture: National Examination Preparation (Physics)** | (2009) Sarawittaya School, Bangkok, Thailand
- **Technical Efficiency of University Hospitals in Thailand** | Bhirombhakdi (2008) Thesis, Chulalongkorn University, Bangkok, Thailand. [Link](#)